

Assignment No. 1

EECS 658

Introduction to Machine Learning

Due: 11:59 PM, Thursday, August 28, 2025

Submit deliverables in a single zip file to Canvas

Files in other formats (e.g., .tar) will not be graded

Name of the zip file: FirstnameLastname_Assignment1 (with your first and last name)

Name of the Assignment folder within the zip file: FirstnameLastname_Assignment1

Deliverables:

1. Copy of Rubric1.docx with your name and ID filled out (do not submit a PDF)
2. Python source code for CheckVersions
3. Screen print showing the successful execution of CheckVersions or copy and paste the output from a console screen to a Word document and PDF it. A screen print is required. Do not expect the grader to run your code.
4. Python source code for NBClassifier
5. Screen print showing the successful execution of NBClassifier
6. Answer and calculations to the following questions:
 - a. Using the confusion matrix, manually calculate the Accuracy value. Does it match the value calculated by your program? If not, why? (Manually includes using a spreadsheet).
 - b. Using the confusion matrix, manually calculate the Precision values for each iris variety. Do they match the values calculated by your program? If not, why?
 - c. Using the confusion matrix, manually calculate the Recall values for each iris variety. Do they match the values calculated by your program? If not, why?
 - d. Using the confusion matrix, manually calculate the F1 values for each iris variety. Do they match the values calculated by your program? If not, why?

Assignment:

- Install Python on your system if it is not already.
 - See “Python for Windows Primer” on Canvas (under Assignment 1) for help on Windows
 - For help on Linux, see:
 - https://wiki.ittc.ku.edu/itc_wiki/index.php/EECS168:SSH_Instructions
 - Virtual Box: <https://www.virtualbox.org/wiki/Downloads>
 - Ubuntu install: <https://ubuntu.com/download/desktop>
 - See “Beginner’s Python Cheat Sheet” on Canvas (under Assignment 1) for help with Python.
- Install the following Python libraries.
 - scipy
 - numpy
 - pandas

- sklearn
- The scipy installation page provides excellent instructions for installing the above libraries on multiple different platforms, such as Linux, mac OS X and Windows. If you have any doubts or questions, refer to this guide, it has been followed by thousands of people.
- To verify you have installed Python and the SciPy libraries write a Python program called CheckVersions that 1) prints out the versions of Python, scipy, numpy, pandas, and sklearn and 2) prints out “Hello World!”
 - Hint: use this code for part 1):


```
# Python version
import sys
print('Python: {}'.format(sys.version))
# scipy
import scipy
print('scipy: {}'.format(scipy.__version__))
# numpy
import numpy
print('numpy: {}'.format(numpy.__version__))
# pandas
import pandas
print('pandas: {}'.format(pandas.__version__))
# scikit-learn
import sklearn
print('sklearn: {}'.format(sklearn.__version__))
```
- Write a Python program called NBClassifier that does the following:
 - Uses 2-fold cross-validation to produce a test set of 150 samples of the iris data set with the Naïve Baysian (GaussianNB) ML models, like was explained in the Supervised Learning lecture. The test set was called “predicted” in the Supervised Learning lecture example.
 - Remember 2-fold cross-validation involves:
 - Dividing the data set into 2 folds
 - Training the model with fold 1
 - Testing the model with fold 2
 - Training the model with fold 2
 - Testing the model with fold 1
 - Concatenating the test results from the 2 folds to get a test set of 150 samples.
 - Prints out the overall accuracy of the classifier.
 - Prints out the confusion matrix.
 - If the values in your confusion matrix does not add up to 150, then you did something wrong.
 - Prints out the P, R, and F1 score for each of the 3 varieties of iris.
 - You may (and probably should) use the Python built-in programs.

Rubric for Program Comments		
Exceeds Expectations (90-100%)	Meets Expectations (80-89%)	Unsatisfactory (0-79%)
Software is adequately commented with prologue comments, comments summarizing major blocks of code, and comments on every line.	Prologue comments are present but missing some items or some major blocks of code are not commented or there are inadequate comments on each line.	Prologue comments are missing all together or there are no comments on major blocks of code or there are very few comments on each line.

Adequate Prologue Comments:

- Name of program contained in the file (e.g., EECS 658 Assignment 1)
- Brief description of the program, e.g.,
 - Check versions of Python & create ML “Hello World!” program
- Inputs (e.g., none, for a function, it would be the parameters passed to it)
- Output, e.g.,
 - Prints out the versions of Python, scipy, numpy, pandas, and sklearn
 - Prints out “Hello World!”
 - Prints out the overall accuracy of the classifier.
 - Prints out the confusion matrix.
 - Prints out the P, R, and F1 score for each of the 3 varieties of iris.
- All collaborators
- Other sources for the code ChatGPT, stackOverflow, etc.
- Author’s full name
- Creation date: The date you first create the file, i.e., the date you write this comment

Adequate comments summarizing major blocks of code and comments on every line:

- Provide comments that explain what each line of code is doing.
- You may comment each line of code (e.g., using `//`) and/or provide a multi-line comment (e.g., using `/*` and `*/`) that explains what a group of lines does.
- Multi-line comments should be detailed enough that it is clear what each line of code is doing.
- Each block of code must indicate whether you authored the code, you obtained it from one of the sources listed in the prolog, or one of your collaborators authored the code, or if it was a combination of all of these.

Collaboration and other sources for code:

- When you collaborate with other students or use other sources for the code (e.g., ChatGPT, stackOverflow):
 - Your comments must be significantly different from your collaborators.
 - More scrutiny will be applied to grading your comments in particular explaining the code “in your own words”, not the source’s comments (e.g., ChatGPT’s comments).

- Failure to identify collaborators or other sources of code will not only result in a 0 on the assignment but will be considered an act of Academic Misconduct.
- Students who violate conduct policies will be subject to severe penalties, up through and including dismissal from the School of Engineering.